

Computational Fluid Dynamics - 12/9/2023

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Exercise 1. We want to simulate the motion of a fluid flowing through the duct represented in Figure 1. The problem is governed by the steady incompressible Navier-Stokes equations. We consider the following boundary conditions:

- a uniform velocity profile is imposed on the inflow boundary Γ_{in} : $\mathbf{u} = \mathbf{i}$;
- a no-slip condition is imposed on the solid wall Γ_{wall} : $\mathbf{u} = \mathbf{0}$;
- a homogeneous Neumann condition is imposed on the outflow boundary Γ_{out} : $\nu \frac{\partial \mathbf{u}}{\partial \mathbf{n}} - p \mathbf{n} = \mathbf{0}$;
- a symmetry condition is imposed on Γ_{symm} : $u_y = 0$, $\frac{\partial u_x}{\partial \mathbf{n}} = 0$.

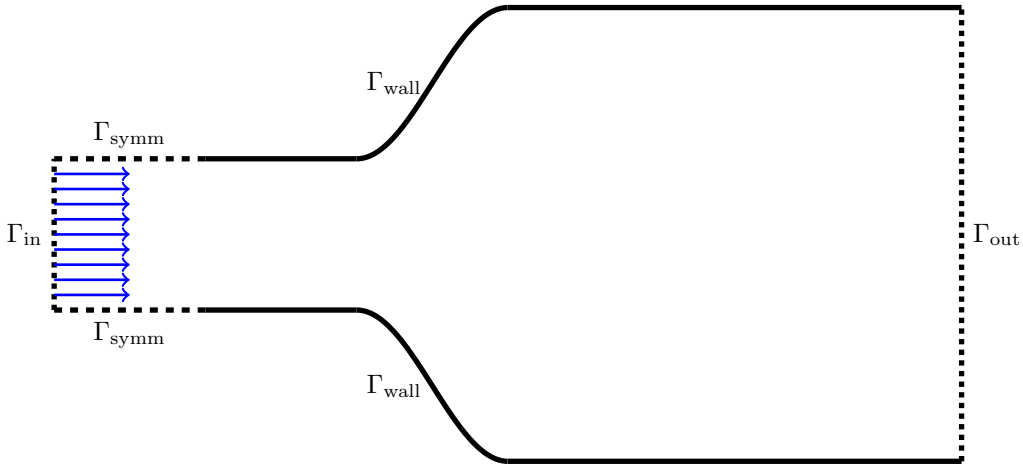


Figure 1: Computational domain

- a) Introduce the Newton method for the solution of the problem in strong form.
- b) Write the weak formulation of the linear problem to be solved at each iteration of the Newton method detailing the choice of the functional spaces and the imposition of the boundary conditions.
- c) Introduce the pressure matrix method (also known as Schur complement method) for the solution of the algebraic system obtained at each iteration and define a suitable preconditioner for its solution by an iterative solver.
- d) Starting from the template `exam.edp`, write a `FreeFEM++` script for the solution of the problem using P_2/P_1 finite elements for velocity and pressure. Consider null velocity and pressure as initial guess and employ the following stopping criterion on the solution increment for the Newton iteration

$$\|\mathbf{u}_k - \mathbf{u}_{k-1}\|_{H^1} + \|p_k - p_{k-1}\|_{L^2} \leq 10^{-3}.$$

Use a tolerance of 10^{-6} for the iterative linear system solution.

- e) Solve the problem for different values of the kinematic viscosity $\nu = 0.1, 0.05, 0.025$, report the maximum number of GMRES iterations required for the solution of the pressure system and the number of Newton iterations required to reach convergence. Comment the results.
- f) Estimate the distance between the reattachment point of the recirculation bubbles and the outflow for the different values of viscosity.